

Document 8: Centralized Session Validation Specification

8.1 Verification Lifecycle Constraints

To preserve real-time audio thread performance, licensing verification is decoupled from the digital signal processing loop. The validation request executes strictly on a dedicated background network thread during plugin initialization or when a user explicitly submits credentials. The audio loop queries a thread-safe local boolean flag to determine authorization status.

8.2 Non-Hardware-Locked Concurrent Session Architecture

The system prevents simultaneous multi-device account sharing without enforcing strict hardware identifier locks. This allows users to move between studio machines freely while restricting execution to one active device at a time.

- **User Token Generation:** Upon purchase, the central licensing server generates a unique alphanumeric license key associated with the user's account record.
- **The Heartbeat Exchange:** Upon startup, the plugin establishes an outbound secure network connection to a lightweight web script endpoint, passing the license key and a newly generated, ephemeral Session ID.
- **Server-Side State Resolution:** The receiving server checks the key against an active session database table:
 - If the key has no active session, the server registers the current Session ID and returns a success token.
 - If the key is already linked to an active session with a different Session ID, the server automatically deletes the older session record, assigns the slot to the new Session ID, and returns an override success token.

8.3 Heartbeat Timeout and Eviction Loop

To gracefully handle cases where a user abruptly closes their host DAW without a clean logout sequence, the plugin sends a lightweight verification ping to the server every 5 minutes.

If the central server receives a ping from a different machine using a newer Session ID, the next scheduled ping from the older machine will receive a rejection payload. Upon receiving a rejection payload, the older plugin instance clears its authorization flag.

8.4 Evaluation Failure States (Demo Mode)

If the background network thread receives an explicit rejection payload, or if no valid key is present, the local authorization flag evaluates to false, forcing the plugin into a restricted demo state:

- **Audio Muting:** Every 60 seconds, the real-time audio thread suppresses output processing for a continuous duration of 3 seconds.
- **Compilation Interruption:** The graphical canvas viewport functions normally, but the background compiler thread is blocked from processing new pixel grids until authorization succeeds.